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• 20160203117

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of	:	GOKARAJU et al.
For	:	ANTI-ADIPOGENIC COMPOSITIONS CONTAINING *PIPER BETLE* AND *DOLICHOS BIFLORUS*
Serial No.:	:	12/679,826
Filed	:	March 24, 2010
Art Unit	:	1655
Examiner	:	Amy Lynn CLARK
Att. Docket	:	LIX 3027
Confirmation No.	:	9829

AMENDMENT AFTER FINAL

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Office Action dated April 26, 2012, please amend the above-identified application as set forth below.

CLAIM AMENDMENTS begin on page 2 of this paper.

REMARKS/ARGUMENTS begin on page 14 of this paper.

CLAIM AMENDMENTS

This listing of claims will replace all prior versions and listings of claims in the application.

Listing of Claims

1-22. (Cancelled)

23. (Currently amended) Anti-adipogenic herbal compositions for inhibiting or preventing or ameliorating disease conditions associated with adipogenesis in mammals, comprising:

from 33% to 67% by weight of a *Piper betle* extract, wherein the *Piper betle* extract is obtained by extraction of *Piper betle* with a solvent selected from the group consisting of ~~water, a~~ at least one polar organic solvent and a combination of a polar organic solvent and water, ~~and mixtures thereof~~ and

an effective amount of a *Dolichos biflorus* extract, wherein the *Dolichos biflorus* extract is obtained by extraction of *Dolichos biflorus* with a solvent selected from the group consisting of ~~water, a~~ at least one polar organic solvent and a combination of a polar organic solvent and water, ~~and mixtures thereof~~;

optionally in combination with at least one biologically acceptable excipient.

24. (Currently amended) The anti-adipogenic compositions as claimed in claim 58, wherein the effective amount of said *Piper betle* extract in the compositions <u>is</u> in the range of 1% to 99% by weight.

25. (Previously presented) The anti-adipogenic compositions as claimed in claim 58, wherein the composition comprises *Piper betle* extract in the range of 1% - 99% by weight and *Dolichos biflorus* extract in the range of from greater than 0% to 70% by weight.

26. (Previously presented) Anti-adipogenic herbal compositions according to claim 23,

wherein the composition comprises *Piper betle* extract in the range of 40% - 60% by weight and *Dolichos biflorus* extract in the range of 40% - 60% by weight.

27. (Currently amended) Anti-adipogenic herbal compositions for inhibiting or preventing or ameliorating disease conditions associated with adipogenesis in mammals, <u>consisting essentially of</u> ~~comprising~~:

an effective amount of a *Piper betle* extract, wherein the *Piper betle* extract is obtained by extraction of *Piper betle* with a solvent selected from the group consisting of water, a polar organic solvent, and mixtures thereof;

optionally an effective amount of a *Dolichos biflorus* extract, wherein the *Dolichos biflorus* extract is obtained by extraction of *Dolichos biflorus* with a solvent selected from the group consisting of water, a polar organic solvent, and mixtures thereof;

optionally in combination with at least one biologically acceptable excipient;

wherein the said *Piper betle* extract further comprises one or more anti-adipogenic or anti-obesic agents selected from the group consisting of extracts of *Commiphora mukul, Boerhavia diffusa, Tribulus terrestris,* ~~and~~ *Zingiber officinale*, and mixtures thereof; and

optionally contains a bio-enhancing agent or a bio-protecting agent.

28. (Previously Presented) The anti-adipogenic compositions as in claim 27, wherein the composition comprises extracts of Piper betle, *Commiphora mukul, Boerhavia diffusa, Tribulus terrestris, Zingiber officinale* along with said bio-enhancing agent, wherein said bio-enhancing agent comprises an extract of *Piper nigrum*.

29. (Previously Presented) The anti-adipogenic compositions as in claim 27, wherein the composition comprises *Piper betle* extract in the range of 10-40%, *Commiphora mukul* extract in the range of 10-40%, *Boerhavia diffusa* extract in the range of 10-40%, *Tribulus terrestris* extract in the range of 10-40%, *Zingiber officinale* extract in the range of 10 - 40% and said extract of *Piper nigrum* in the range of 10-40%.

30. (Previously Presented) The anti-adipogenic compositions as in claim 27, wherein the said compositions optionally comprise anti-adipogenic or anti-obese agents selected from the group consisting of an *Holoptelia integrifolia* extract, a *Garcinia cambogia* extract, a green tea extract, a green coffee bean extract, an eucalyptus plant extract, a bitter orange (*Citrus aurantium*) extract, Conjugated linoleic acid, a *Hoodia Gordonii* extract, an *Allium sativum* extract, chromium (III) complexes, DHEA, and 7-Keto-DHEA..

31. (Currently amended) The anti-adipogenic compositions as claimed in claim 27, wherein said extracts are formulated optionally with biologically acceptable excipients into specific dosage forms selected from the group consisting of nutraceuticals, pharmaceuticals, and dietary forms.

32. (Currently amended) The anti-adipogenic compositions as in claim 27, wherein said compositions further comprise an effective amount of at least one pharmaceutically or nutritionally or dietetically acceptable antioxidant, adaptogen, anti-inflammatory agent, anti-diabetic agent, bio-protectant, trace metals or mixture thereof to form a formulation.

33. (Currently amended) The anti-adipogenic compositions as claimed in claim 27, wherein said composition is in a pharmaceutically or nutraceutically suitable form for oral administration, said pharmaceutically or nutraceutically suitable form being selected from the group consisting of tablets, soft capsules, hard capsules, pills, granules, powders, emulsions, suspensions, syrups, and pellets.

34. (Previously Presented) The anti-adipogenic compositions as claimed in claim 31, wherein the pharmaceutically or nutraceutically suitable form is suitable for parenteral administration selected from the group consisting of injections, drops, and suppositories.

35. (Currently amended) The anti-adipogenic compositions as claimed in claim 27, wherein said compositions are in the form of a dietary formulation.

36. (Previously Presented) A dosage form comprising a composition of claim 28 and a sufficient amount of an excipient.

37. (Canceled)

38. (Canceled)

39. (Previously Presented) The anti-adipogenic composition as claimed in claim 28, wherein said composition further comprises a pharmaceutically or nutraceutically or dietetically or veterinary acceptable additive, wherein the said additive is a binder, diluent, antioxidant or a lubricant.

40. (Currently amended) The anti-adipogenic compositions as claimed in claim 23, wherein a source of the *Piper betle* extract is a plant part selected from the group consisting of leaf, seed, trunk, root and combinations thereof; and

a source of the *Dolichos biflorus* extract is a plant part selected from the group consisting of leaf, seed, trunk, root and combinations thereof.

41. (Currently amended) The anti-adipogenic compositions as claimed in claim 23, wherein the *Piper betle* extract and the *Dolichos biflorus* extract are derived by extraction from plant parts with a solvent selected from the group consisting of water, a polar organic solvent, and a mixture thereof, wherein the solvent used for extraction of the plant parts is selected from the group consisting of methanol, ethanol, propanol, isopropanol, n-butanol, isobutanol, sec-butanol, tert-butanol, dimethyl ketone, methyl ethyl ketone, acetone, methyl isobutyl ketone, water and mixtures thereof.

42. (Withdrawn) A method for inhibiting adipogenesis and fat accumulation in a mammal, comprising administering to a said mammal an effective amount of a composition as claimed in claim 28, as a dietary or nutraceutical or pharmaceutical composition.

43. (Withdrawn) A method for inhibiting or preventing or treating adipogenesis mediated diseases in a mammal comprising administering to a said mammal an effective amount of a composition as claimed in claim 28.

44. (Withdrawn) A weight controlling treatment regimen, comprising administering the anti-adipogenic compositions as claimed in claim 23, wherein the compositions are administered either before meals or as a meal replacement as a part of an overall weight control plan.

45. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein the said compositions optionally comprise anti-adipogenic or anti-obese agents selected from the group consisting of an *Holoptelia integrifolia* extract, a *Garcinia cambogia* extract, a green tea extract, a green coffee bean extract, a eucalyptus plant extract, a bitter orange (*Citrus aurantium*) extract, Conjugated linoleic acid, a *Hoodia gordonii* extract, an *Allium sativum* extract, chromium (III) complexes, DHEA, and 7-Keto-DHEA.

46. (Currently amended) The anti-adipogenic compositions as claimed in claim 23, wherein said extracts are formulated with biologically acceptable excipients into specific dosage forms, said dosage forms being selected from the group consisting of nutraceuticals, pharmaceuticals, or dietary forms.

47. (Previously Presented) The anti-adipogenic compositions as claimed in claim 32, wherein said bio-protectant is *Curcuma longa.*

48. (Currently amended) The anti-adipogenic compositions as claimed in claim 27, wherein said compositions are in the form of a dietary formulation selected from the group consisting of chocolate bars, nutritional bars, creams, jams, gels, beverages selected from the group consisting of coffee, tea, a milk-containing beverage, a lactic acid bacteria-containing beverage, drops, candies, chewing gums, chocolates, gummy candies, yoghurt, ice cream, puddings, soft adzuki-bean jellies, jellies, and cookies.

49. (Previously Presented) A pharmaceutically acceptable dosage form comprising a composition of claim 23, further comprising a sufficient amount of a pharmaceutically acceptable excipient.

50. (Canceled)

51. (Canceled)

52. (Withdrawn) A method for inhibiting or preventing or treating adipogenesis mediated diseases in a mammal comprising administering to a said mammal an effective amount of a composition as claimed in claim 23.

53. (Withdrawn) A method for inhibiting adipogenesis and fat accumulation in a mammal, comprising administering to a said mammal an effective amount of a composition as claimed in claim 23, as a dietary or nutraceutical or pharmaceutical composition.

54. (Withdrawn) A method for inhibiting or preventing or treating adipogenesis mediated diseases in a mammal comprising administering to a said mammal an effective amount of a composition as claimed in claim 26.

55. (Withdrawn) The method according to claim 54, wherein said adipogenesis mediated diseases are obesity, excess body weight, fat accumulation, or a combination thereof.

56. (canceled)

57. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein said compositions are in the form of a dietary formulation selected from the group consisting of chocolate bars, nutritional bars, creams, jams, gels, beverages selected from the group consisting of coffee, tea, a milk-containing beverage, a lactic acid bacteria-containing beverage, drops, candies, chewing gums, chocolates, gummy candies, yoghurt, ice cream, puddings, soft adzuki-bean jellies, jellies, and cookies.

58. (Previously presented) An anti-adipogenic herbal composition for inhibiting disease conditions associated with adipogenesis in mammals, comprising:

an effective amount of a *Piper betle* extract, wherein the *Piper betle* extract is obtained by extraction of *Piper betle* with water or a polar organic solvent or a mixture thereof;

an effective amount of a *Dolichos biflorus* extract, wherein the *Dolichos biflorus* extract is obtained by extraction of *Dolichos biflorus* with water or a polar organic solvent or a mixture thereof; and

an optional biologically acceptable excipient;

wherein said *Piper betle* extract and said *Dolichos biflorus* extract are the only herbal ingredients present in said anti-adipogenic herbal composition.

59. (Currently amended) An anti-adipogenic herbal composition for inhibiting disease conditions associated with adipogenesis in mammals, consisting essentially of ~~comprising~~ a first herbal component and an optional biologically acceptable excipient;

said first herbal component comprising an effective amount of *Piper betle* extract obtained by extraction of *Piper betle* with at least one polar organic solvent or a combination of a polar organic solvent and water, ~~water or a polar organic solvent or a mixture thereof~~ and an effective amount of *Dolichos biflorus* extract obtained by extraction of *Dolichos biflorus* with at least one polar organic solvent or a combination of a polar organic solvent and water ~~water or a polar organic solvent or a mixture thereof~~;

wherein said first herbal component is prepared by a process consisting essentially of mixing said *Piper betle* extract and said *Dolichos biflorus* extract.

60. (Previously presented) The anti-adipogenic compositions as claimed in claim 23, further comprising a second herbal component selected from the group consisting of an *Holoptelia integrifolia* extract, a *Garcinia cambogia* extract, a green tea extract, a green coffee bean extract, a eucalyptus plant extract, a bitter orange (*Citrus aurantium*) extract, a *Hoodia gordonii* extract, an *Allium sativum* extract, and mixtures thereof.

61. (Currently amended) The anti-adipogenic compositions as claimed in claim 23, wherein:

the *Piper betle* extract is obtained by extraction of *Piper betle* with ~~water or~~ either ~~a~~ at least one C1-C4 alcohol ~~or a mixture thereof~~, or a mixture of water and at least one C1-C4 alcohol; and

the *Dolichos biflorus* extract is obtained by extraction of *Dolichos biflorus* with ~~water or~~ either ~~a~~ at least one C1-C4 alcohol ~~or a mixture thereof~~, or a mixture of water and at least one C1-C4 alcohol.

62. (Previously presented) The anti-adipogenic compositions as claimed in claim 23, wherein:

the *Piper betle* extract is obtained by extraction of *Piper betle* leaves; and

the *Dolichos biflorus* extract is obtained by extraction of *Dolichos biflorus* leaves.

63. (Currently amended) The anti-adipogenic compositions as claimed in claim 23, wherein:

the *Piper betle* extract is obtained by extraction of *Piper betle* leaves with ~~water or~~ either ~~a~~ at least one C1-C4 alcohol ~~or a mixture thereof~~, or a mixture of water and at least one C1-C4 alcohol; and the *Dolichos biflorus* extract is obtained by extraction of *Dolichos biflorus* leaves with ~~water or~~ either ~~a~~ at least one C1-C4 alcohol ~~or a mixture thereof~~, or a mixture of water and at least one C1-C4 alcohol.

REMARKS/ARGUMENTS

Claims 23-63 are pending in this application. By this Amendment, claims 61-63 are added; claims 23-24, 26-27, 30-33, 35, 36, 40-41, 48 and 58-59 are amended; and claims 37-38, 50-51 and 56 are canceled. Claims 42-44 and 52-55 have been withdrawn. Claims 61-63 are added.

REJECTIONS UNDER 35 U.S.C. 102

Claims 23-27, 30-36, 40, 41, 46, 48, 49, 57, 59, 60 and 61 are rejected under 35 U.S.C. 103(a) as being unpatentable over the TKDL reference Hdrogepathya. According to the Examiner,

> Hdrogepathya teaches a therapeutic formulation (dietary formulation) for treating heart disease (which reads on treating a condition associated with adipogenesis) consisting of Dolichos biflorus, Piper betle, rain water, buttermilk (which reads on a milk containing beverage and biologically acceptable excipient), Allium sativum (which reads on a bio-protectant) and Zingiber officinale (which reads on an anti-inflammatory agent). Please note that combining water with Dolichos biflorus and Piper betle in a composition would provide a water extract of Dolichos biflorus and Piper betle.

This rejection has been addressed by amending claims 23 and 57 to recite use of a polar organic solvent or a mixture of water and a polar organic solvent as an extraction solvent. Hdrogepathya does not show extraction solvents other than water. Accordingly, claims 23 and 57 and their dependent claims are allowable over Hdrogepathya.

Claim 27 has been amended to change "comprising" to "consisting essentially of." The transitional phrase "consisting essentially of" limits the scope of a claim to the specified materials or steps "and those that do not *materially* affect the *basic* and *novel* characteristic(s)" of the claimed invention. *In re Herz*, 537 F.2d 549, 551-52, (CCPA 1976). The claimed composition is an *herbal composition.* Hdrogepathya requires non-herbal ingredients, such as meat juices from animals and birds, as ingredients. Meat juices are obtained from animal sources, and not herbal sources. The current disclosure consistently refers to the composition as an herbal composition, and does not describe the use of animal-derived components. Accordingly, meat juices are not encompassed by the language "consisting essentially of," as applied to a herbal composition.

Claims 23-36, 39-41, 45-49, 57 and 59-61 were rejected under 35 U.S.C. 103(a) as being unpatentable over Hdrogepathya (V), in view of "WorldHealth.net Tribulus Terrestris" (W), Oudhia (X), and Sharma and/or Mohanty. However, "WorldHealth.net Tribulus Terrestris" (W), Oudhia (X), and Sharma and/or Mohanty were relied upon to teach specific herbal ingredients other than Dolichos biflorus and Piper betle, which are not recited by Hdrogepathya. These references do not overcome the deficiency in Hdrogepathya, as they do not show preparation of extracts of Dolichos biflorus and Piper betle using solvents other than water. Additionally, they fail to provide any motivation to remove non-herbal ingredients from Hdrogepathya.

Claims 23-26, 40, 41,46,49,57,59, and 61-63 were rejected under 35 U.S.C. 103(a) as being unpatentable over Arambewela et al., in view of the TKDL reference Dawa E-Kulthi Barae Ziyabitus. The Examiner argues that Arambewela teaches that aqueous and ethanol extracts of leaves of Piper betel have antidiabetic activity and can be administered orally. Dawa E-Kulthi Barae Ziyabitus teaches a decoction of the whole plant of Dolichos biflorus for treating diabetes. The Examiner argues that it would have been obvious to combine these references since both are useful for treating diabetes. However, decoction is understood to be a method of extraction, by boiling, of herbal or plant material. Decoction involves first mashing, and then boiling in water to extract organic substances. Accordingly, the decoction of Dawa E-Kulthi Barae Ziyabitus is a water extract. Dawa E-Kulthi Barae Ziyabitus does not teach or suggest use of a polar organic solvent or a mixture of water and a polar organic solvent as an extraction solvent.

Additionally, Dawa E-Kulthi Barae Ziyabitus does not provide an amount of Dolichos biflorus as an active ingredient, merely reciting an amount of "q.s." Dawa E-Kulthi Barae Ziyabitus is silent as to the concentration of active ingredient in the decoction, or the amount of decoction in the composition. Therefore, since Dawa E-Kulthi Barae Ziyabitus does not teach how much of the active ingredient Dolichos biflorus is present in the composition, Dawa E-Kulthi Barae Ziyabitus is not an enabling reference. There is no teaching or suggestion as to an amount of Dolichos biflorus which is effective in the treatment of diabetes. Accordingly, a person of ordinary skill in the art would have been unable to combine Arambewela et al. and the TKDL reference Dawa E-Kulthi Barae Ziyabitus to produce an effective composition in the absence of undue experimentation as to amounts of Dolichos biflorus.

Claims 23-26, 40, 41,46,49 and 57-61 were rejected under 35 U.S.C. 103(a) as being unpatentable over Arambewela et al., in view of Neelakantan (US 5916567).

Neelakantan teaches an anti-diabetic therapeutic product containing powdered Dolichos biflorus seeds in an amount of from 89.5-98.5% powdered inner seed and 1.5-10.% powdered outer shell of Dolichos biflorus seeds. The composition of Neelakantan does not include a solvent extract, as recited in the current claims. The composition of Neelakantan contains fibers, where "[t]he fibers of the mixture makes a sheath in the intestine to reduce the absorption of the carbohydrates and natural insulin into the blood stream." A solvent extract does not include fibers, and is unable to function in the same manner as the powdered seeds of Neelakantan. The Examiner makes the argument that the combination of an aqueous or ethanol extract of Piper betel leaves with Dolichos biflorus would provide an aqueous or ethanol extract of Dolichos biflorus. Accordingly, the Examiner is not relying on the composition of Neelakantan itself, but a hypothetical composition obtained by extraction of the composition of Nellakantan with a solvent.

However, even if this combination did provide a solvent extract of Dolichos biflorus, the resulting solvent extract would not include fibers from the powdered Dolichos biflorus seeds of Neelakantan. Since Neelakantan teaches that these fibers are required to "make[] a sheath in the intestine to reduce the absorption of the carbohydrates and natural insulin into the blood stream," the loss of these fibers by solvent extraction of the composition of Neelakantan would render the composition of Neelakantan ineffective for its stated purpose. If a proposed modification would render the prior art invention being modified unsatisfactory for its intended purpose, then there is no suggestion or motivation to make the proposed modification. *In re Gordon*, 733 F.2d 900, 221 USPQ 1125 (Fed. Cir. 1984).

Further, the Examiner agreed in an interview dated December 21, 2011, that the powdered Dolichos biflorus seeds of Neelakantan are not obtained by solvent extraction. The Examiner agreed in the interview that amendment of the claims to specify a Dolichos biflorus extract obtained by extraction with a solvent would overcome a rejection using Neelakantan as a reference.

CONCLUSION

While we believe that the instant amendment places the application in condition for allowance, should the Examiner have any further comments or suggestions, it is respectfully requested that the Examiner telephone the undersigned attorney in order to expeditiously resolve any outstanding issues.

In the event that the fees submitted prove to be insufficient in connection with the filing of this paper, please charge our Deposit Account Number 50-0578 and please credit any excess fees to such Deposit Account.

Respectfully submitted,
KRAMER & AMADO, P.C.

Date: June 26, 2012

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